Computer Science — Class 12 (PCTB, English)

Solved Exercises | Punjab Curriculum and Textbook Board

Note: All 14 chapters are now complete in the book's own order (Chapter 2, Basic Concepts and Terminology of Databases, was missing from the old notes and is added here). Each chapter's old short selection has been replaced by its full exercise set, solved from the book (objective answers follow the book's own Answers key, pp. 240-242). The Appendix preserves the old Arrays items, flagged: this book has NO Arrays chapter in its Contents.

Chapter 1: Data Basics

Exercise 1c — Question 1 (Fill in the blanks — book key, p. 240)

DBMS stands for Database Management System .

A record is a collection of related fields.

A file is a collection of records .

Before processing the data is recorded in transaction files .

A database is a collection of logically related data.

The data definitions is stored in data dictionary .

SQL stands for Structured Query Language .

Hierarchical data model has the general shape of a(n) Organizational Chart .

Data is a collection of facts, figures, statistics .

Processed data is called Information .

Exercise 1c — Question 2 (Multiple choice — book key, p. 240)

Which of the following represents a collection of concepts that are used to describe the structure of a database?

A. data warehouse

✓ B. data model

C. data structure

D. data type

Which of the following data model is more flexible?

A. Network data model

B. Hierarchical data model

C. Relational data model

✓ D. object data model

Which of the following type of file require largest processing time?

✓ A. Sequential file

B. Random file

C. Indexed sequential file

D. Direct access file

Which of the following may be a temporary file?

A. Master file

✓ B. Transaction file

C. Backup file

D. None of these

SQL is a(n):

A. Unstructured language

✓ B. Structured language

C. Object oriented language

D. Software

Exercise 1c — Question 3 (True / False — book key, p. 240)

Data can only be processed through computers

False

The traditional file system approach has many advantages over DBMS approach

False

Data dictionary is used to view the meanings of database terminology

False

Master file is the latest updated file which never becomes empty, ever since it is created

True

SQL is used to retrieve information from the database based on certain criteria

True

The Network Data Model is more popular and widely used than Relational Data Model

False

Indexed sequential files can be processed sequentially as well as randomly

True

Backup files store data prior to its processing

False

Microsoft ACCESS is a relational database management system

True

A report generator is used to produce a printed document from the database

True

Exercise 1c — Questions 5-14 (Descriptive)

Differentiate between Data and Information.

Data is a collection of raw facts, figures and statistics, such as marks of students.

Information is processed data that is meaningful and useful for decision making, such as the result card prepared from those marks.

Data alone has no meaning, while information gives meaning to data.

What activities are involved in data processing? Discuss in details.

Data processing involves capturing data from its source and recording it in transaction files.

Then the data is manipulated: it is classified, sorted, calculated and summarized.

Finally the processed result is stored and presented as information, and reports are produced from it.

Define file, record and field in details.

A field is a single piece of information about a person or thing, such as a student's name.

A record is a collection of related fields, such as one student's complete row of data.

A file is a collection of related records, such as the record of all students of a class.

Describe the file types from usage point of view and functional point of view.

From usage point of view, files are master files (permanent, regularly updated), transaction files (temporary, holding new changes) and backup files (copies kept for safety).

From functional point of view, files are sequential files, direct access files and indexed sequential files.

Sequential files are processed record by record, direct files allow direct access to any record, and indexed sequential files allow both sequential and random processing.

How do we organize the files on storage media?

Files are organized on storage media as sequential, direct or indexed sequential files.

In sequential organization records are stored one after another and read in order.

In direct organization each record is reached directly with the help of a key, and indexed sequential organization uses an index to reach records directly or in order.

In general, what activities are to be performed on the databases? Discuss in details.

A database is created by defining its structure and entering data into its tables. Data is then updated, deleted and retrieved whenever needed, and queries are run to get specific information.

Reports are produced from the database, and backup and recovery keep the data safe.

What are the four major components of the database systems? Write in details.

The four major components are data, hardware, software and users.

Data is the stored facts, hardware is the computer system, software is the DBMS that manages the data, and users are the people who use the database.

The DBMS links the users to the data and controls how the data is stored and retrieved.

Discuss the objectives of the databases in your own words.

The objectives of databases are to store data in an organized way and to remove the problems of the traditional file system.

Data is shared by many users, duplication is reduced, and data remains consistent and secure.

Data independence is provided, so changing the storage structure does not disturb the programs using the data.

Describe the different database models.

The hierarchical data model organizes data in a tree-like structure with a parent and children.

The network data model allows a record to have more than one parent, forming a network of records.

The relational data model stores data in tables, and the object data model stores

data as objects with attributes and methods.

Discuss the objectives and features of the DBMS.

A DBMS manages a database: it stores data, controls access to it and keeps it consistent and secure.

Its features include a data dictionary, utilities, a query language, a report generator, access security and backup and recovery.

Its objectives are data sharing, less duplication, data independence, integrity and security of data.

What are the advantages and disadvantages of the DBMS?

Advantages of a DBMS are shared data, reduced duplication, data consistency, security and independence of data from programs.

Disadvantages are extra system overhead, the need for trained staff, and problems in selecting the wrong database environment.

A DBMS also needs extra hardware and software, which increases the cost of the system.

Chapter 2: Basic Concepts and Terminology of Databases

Exercise 2c — Question 1 (Fill in the blanks — book key, p. 240)

A(n) Table is a two dimensional array containing descriptive information about an entity.

A(n) Entity is any thing about which the information is kept in the database.

In a table the order of rows and columns is Insignificant .

In a relation, the attribute or a combination of attributes that uniquely identifies a record is called Primary key .

A(n) Attribute describes the characteristics of an entity.

A(n) Foreign key is an attribute in a table whose values must match a primary key in another table.

A(n) DBA is responsible for the design, implementation, operation, management and maintenance of the database.

A(n) Composite key consists of two or more attributes.

A(n) View is the dynamic result of one or more relational operations on the base relations to produce another relation.

The data refers to raw facts and figures.

Exercise 2c — Question 2 (Multiple choice — book key, p. 240)

Insert command is used to insert:

A. a new table

✓ B. a new record

C. a view

D. dependencies

CREATE command is used to create a:

✓ A. table

B. view

C. report

D. query

SQL is used for:

A. data definition

B. data manipulation

✓ C. data definition and manipulation

D. searching records

The foreign key is found in:

A. parent table

✓ B. dependant table

C. pivot table

D. index table

A table must have:

✓ A. primary key

B. secondary key

C. composite key

D. sort key

Exercise 2c — Question 3 (True / False — book key, p. 240)

The view is not stored in the database

True

Two tables can not have the same name in the database

True

Index makes the searching of a record faster

True

Secondary key must be unique

False

The primary key can not work as a sort key

False

The DBA is responsible for maintaining the database

True

A file is a collection of related fields

False

DBMS provide more security to protect data than traditional file management systems

True

DBMS is a software used to train database administrator

False

A relation is also termed as a tuple in relational database

False

Exercise 2c — Questions 4-10 (Descriptive)

How the Records and Files are constructed in traditional File Management System?

In the traditional file management system, data is stored in separate files for each application.

Each file is made of records, and each record is made of related fields.

Programs are written separately for each file, so the same data is often repeated in many files.

How the data is stored and retrieved in FMS (file management system)?

In a file management system, data is stored in files on a storage device in the form of records.

To retrieve data, a program reads the file record by record until the required record is found.

There is no central control, so searching and updating data is slow and difficult.

How the tables/relations are formed up in DBMS?

In a DBMS, data is organized into tables, also called relations.

Each table has rows called tuples and columns called attributes, and every row is identified by a primary key.

Related tables are linked through keys, so data is stored without unnecessary duplication.

Discuss the data manipulation in DBMS system.

Data manipulation in a DBMS means inserting, deleting, updating and retrieving data.

The SQL commands INSERT, DELETE, UPDATE and SELECT are used for these operations.

Manipulation is done through queries, so the user can get exactly the required information from the tables.

Write down the properties of relations in details.

A relation is a two dimensional table with rows and columns.

The order of rows and columns is insignificant, and no two rows are exactly alike.

Each column has a single data type, each cell holds a single value, and every row is uniquely identified by a key.

What is a VIEW? How do we create it?

A view is the dynamic result of one or more relational operations on base relations, which produces another relation.

A view is not stored in the database; it is created from tables whenever it is used.

In SQL a view is created with the CREATE VIEW command, giving the view a name and a SELECT statement.

What is usage of indexes in FMS and DBMS?

An index is a table that stores the key values of records with their storage addresses.

An index makes the searching of a record faster, because the record is reached directly instead of reading the whole file.

Indexes are used in both file management systems and DBMS to speed up searching and sorting.

Chapter 3: Data Base Design Process

Exercise 3c — Question 1 (Fill in the blanks — book key, p. 240)

During Requirement analysis phase, the project requirements are gathered and identified.

DFD stands for Data Flow Diagram .

The process of identifying data objects and relationship between them is called Data Modeling .

The number of occurrences of participating entities in a relationship is determined by the Cardinality ratio.

Modality determines whether the participation of an entity in a relationship is Mandatory or optional.

ERD stands for Entity Relationship Diagram .

In ERD model, a(n) Entity is represented by a rectangular box.

In Centralized database systems, all the data is stored at a single site.

In Replicated database multiple copies of the same data are stored at different sites on the network.

In distributed databases, the data is Distributed among various sites.

Exercise 3c — Question 2 (Multiple choice — book key, p. 240)

Which of the following keys does not hold uniqueness property?

A. candidate key

B. foreign key

C. primary key

✓ D. secondary key

An entity related to itself in an ERD model refers to:

✓ A. recursive relationship

B. one-to-many relationship

C. many-to-many relationship

D. one-to-one relationship

Database development process involve mapping of conceptual data model into:

A. Object oriented data model

B. Network data model

✓ C. Implementation model

D. Hierarchical data model

In ERD model, the relationship between two entities is represented by a:

A. diamond symbol

B. rectangular box

C. oval symbol

✓ D. line

In hybrid distribution which kind of fragments are stored at only one site:

A. critical fragments

✓ B. non-critical fragments

C. critical and non-critical fragments

D. only large fragments

Exercise 3c — Question 3 (True / False — book key, p. 240)

In one-to-one relationship only one instance of each entity can participate in the relationship

True

The optional modality is represented by 1

False

One-to-many is a uni-directional relationship

False

In ERD model, a condition is mentioned in a diamond symbol

True

ERD is a physical data model

False

In hybrid distribution the database is partitioned in critical and non-critical fragments

True

In distributed databases, the consistency refers to availability of same data at all sites of the network

True

Indexing maximizes the time required to search a piece of information from a database

False

Analysis is less important activity than coding, so minimum time should be spent over analyzing the system

False

Relationship defines the logical connection between entities

True

Exercise 3c — Questions 4-12 (Descriptive)

Describe different steps involved in analysis stage while designing a database.

The analysis stage starts with a feasibility study, which checks whether the system can be built within time and cost.

Then requirement analysis gathers the needs of the users and the data they want to store.

After that project planning is done and data analysis identifies the data objects and their relationships.

Explain the following with the help of figures: (i) Entity/Object (ii) Attribute (iii) Relationship (iv) Cardinality (v) Modality.

An entity or object is a thing about which information is kept, shown in an ERD by a

rectangular box.

An attribute describes a characteristic of an entity, shown by an oval.

A relationship is the logical connection between entities, shown by a diamond; cardinality shows how many instances take part, and modality shows whether participation is mandatory or optional.

Draw and explain ER diagram for the system of getting admission in your college.

The admission system has entities such as Student, Class and Fee, drawn as rectangular boxes.

The student fills an admission form, which is the relationship between Student and Class, drawn as a diamond.

Attributes of Student are name, roll number and address, and the diagram shows one student taking admission in one class.

Explain the following: (i) Physical data model (ii) Conceptual data model.

A conceptual data model describes what data is needed and how entities are related, without caring about storage.

It is drawn as an ER diagram during the design of a database.

A physical data model describes how the data is actually stored on the storage device, including tables, indexes and file organization.

What are the components of a logical data model?

The components of a logical data model are entities, attributes, relationships, keys and constraints.

Entities are the objects, attributes are their characteristics, and relationships connect the entities.

Keys identify each record, and constraints are the rules that keep the data correct and consistent.

What elements combined, produce the physical database design? Explain.

Physical database design is produced by combining tables, fields, data types,

indexes and file organization.

The logical design is mapped into actual tables, and storage structures are selected for fast access.

Security, backup and recovery arrangements are also decided in the physical design.

Define and explain the following terms: (i) Data distribution strategy (ii) File Organization.

A data distribution strategy decides how data is divided among the sites of a network, such as centralized, replicated or distributed storage.

File organization is the way records are arranged in a file on a storage device.

Common file organizations are sequential, direct and indexed sequential organization.

Define the term Analysis. Briefly discuss the following terms: (i) Feasibility study (ii) Requirement Analysis (iii) Project Planning (iv) Data Analysis.

Analysis is the stage of system development in which the problem is studied and the requirements of the new system are gathered.

A feasibility study checks whether the system is possible, requirement analysis collects user needs, project planning fixes time and cost, and data analysis identifies the data objects and their relationships.

Briefly explain the database design process with the help of a diagram.

The database design process starts with analysis, in which requirements are gathered and feasibility is checked.

Then the conceptual design is made as an ER diagram, which is mapped into a logical design of tables.

Finally the physical design is prepared, tables are created, and the database is implemented and tested.

Chapter 4: Data Integrity and Normalization

Exercise 4c — Question 1 (Fill in the blanks — book key, p. 240)

Entity integrity constraint states that the Primary key can not be null.

A Foreign key must refer to the primary key in another table or it must be null.

Normalization is the process of converting Complex structures into simple and stable structures.

A(n) Functional dependency is a partial relationship between attributes of an entity.

During the first normal form Repeating groups are removed.

To be in 2NF, a relation must be in 1-NF .

In 3NF, no Transitive dependency exists.

A(n) Partial functional dependency exists when one or more non-key attributes are functionally dependant on part of the primary key.

When a new record is added in a relation, it may cause Insertion anomaly anomaly.

Referential integrity is a constraint on Foreign key value.

Exercise 4c — Question 2 (Multiple choice — book key, p. 240)

In 3NF, which form of dependency is removed?

A. functional

B. non-functional

C. associative

✓ D. transitive

In relational database, a table is also called a:

A. tuple

✓ B. relation

C. file

D. schema

In 3NF, a non-key attribute must not depend on a(n):

✓ A. non-key attribute

B. key attribute

C. composite key

D. sort key

Different attributes in two different tables having same name are referred to as:

A. synonym

✓ B. homonym

C. acronym

D. mutually exclusive

Every relation must have a:

✓ A. primary key

B. candidate key

C. secondary key

D. composite key

Exercise 4c — Question 3 (True / False — book key, p. 240)

Normalization is the process of converting complex data structures into simple data structures

True

A relation is decomposed to convert it from 1NF to 2NF

True

The primary key can not be a composite key

False

In 2NF, every non-key attribute must depend on the key attribute

True

A relationship involving three relations is known as a ternary relationship

True

A database anomaly leads the database to an inconsistent state

True

Partial dependencies are removed in 3NF

True

A relation may have multiple primary keys

False

In relational database, no relation can exist in isolation

True

The database is normalized to avoid certain database anomalies

True

Exercise 4c — Questions 4-15 (Descriptive)

What is meant by data integrity? What are the two types?

Data integrity means the correctness and consistency of data stored in a database.

The two types are entity integrity and referential integrity.

Entity integrity requires that the primary key is never null, and referential integrity requires that a foreign key matches a primary key in another table or is null.

What do we do to attain entity integrity?

To attain entity integrity, a primary key is defined for every relation.

The primary key value is never left null and is never repeated in the table.

In this way every record is uniquely identified and the data remains correct.

Define referential integrity. How can it be achieved?

Referential integrity is the rule that a foreign key value must match a primary key value in another table, or it must be null.

It is achieved by defining foreign keys when tables are created and by controlling insertion, deletion and updating of records.

In this way no record refers to a record that does not exist.

Explain the following terms: (i) Synonym (ii) Homonym (iii) Redundancy (iv) Mutual Exclusiveness of data.

A synonym means the same attribute having different names in different tables.

A homonym means different attributes in different tables having the same name.

Redundancy means the same data stored again and again, and mutual exclusiveness means data values that cannot occur together.

What is normalization? How it can be used to bring the database in a consistent state?

Normalization is the process of converting complex data structures into simple and stable structures.

Relations are decomposed into smaller relations to remove repeating groups, partial dependencies and transitive dependencies.

In this way anomalies are removed and the database becomes consistent.

When is a relation in first normal form? Explain with example.

A relation is in first normal form when it has no repeating groups and every cell holds a single value.

For example, if a student record holds many subjects in one field, it is not in first normal form.

When each subject is placed in a separate row, the relation comes into first normal form.

What are the conditions for a relation to be in second normal form? Give example.

A relation is in second normal form when it is in first normal form and every non-key attribute depends on the whole primary key.

Partial functional dependencies, in which a non-key attribute depends on only part of a composite key, are removed.

For example, a table with (Roll No, Subject) as key is split so that subject marks

depend on both, while student name depends only on roll number.

Define transitive dependency. How it can be removed? Explain with the context of normalization.

A transitive dependency exists when a non-key attribute depends on another non-key attribute instead of the key.

For example, if city depends on postal code and postal code depends on the key, then city is transitively dependent.

It is removed by splitting the relation, which brings the relation into third normal form.

What are the database anomalies? Briefly discuss insertion, deletion and modification anomalies.

Database anomalies are the problems that arise when data is stored with redundancy.

An insertion anomaly occurs when a new record cannot be added without adding unrelated data.

A deletion anomaly occurs when deleting one record removes other useful data, and a modification anomaly occurs when the same data must be changed in many places.

What anomalies arises due to transitive dependency? Discuss briefly.

Transitive dependency causes insertion, deletion and modification anomalies.

A new value of a dependent attribute cannot be added without the key value, and deleting a record may remove the dependent data.

Changing the dependent value in one place leaves the other copies unchanged, which makes the data inconsistent.

Define functional dependency? How partial dependencies effect a relation?

A functional dependency is a relationship in which the value of one attribute determines the value of another attribute.

A partial dependency exists when a non-key attribute depends on only part of a composite primary key.

Partial dependencies cause redundancy and anomalies, so they are removed to bring the relation into second normal form.

Convert the ER diagram you have designed in the previous exercise for the admission system of your college to relational database. Also normalize the relations up to third normal form.

The admission ER diagram is converted into relations such as Student (Roll No, Name, Address) and Class (Class ID, Class Name).

The primary keys are underlined, and the foreign key in the admission relation refers to the student and class tables.

Repeating groups are removed for first normal form, partial dependencies for second normal form, and transitive dependencies for third normal form.

Chapter 5: Introduction to Microsoft Access

Exercise 5c — Question 1 (Fill in the blanks — book key, p. 240)

IDE stands for Integrated development environment .

A DBMS is basically a computerized record keeping system.

RDBMS stands for Relational database management system .

The Table object is used to store data in a database.

The Query object is used to retrieve data from a database.

A field with Auto number data type is automatically incremented by Access each time a new record is entered.

Each row of a table is divided into columns called Field .

Each row of a table representing a set of information is called Tuple or Record .

The window that is used to display, enter and edit data on the screen is called Form

.

A database consists of Four major database objects, which are used to store and retrieve data to and from the database.

Exercise 5c — Question 2 (Multiple choice — book key, p. 240)

A database consists of various components called the:

A. Tool

B. Properties

C. Entities

✓ D. Object

Which of the following object of database is used to retrieve data from database?

✓ A. Queries

B. Forms

C. Reports

D. Tables

The output of a query is in the form of a:

✓ A. Table

B. Form

C. Report

D. Query

Which of the following object is used to retrieve data from database and present in a formatted way?

✓ A. Report

B. Form

C. Table

D. Query

Microsoft Access saves the database with the extension:

✓ A. .mdb

B. .msdb

C. .madb

D. None of them

A record is a complete set of ___ field.

A. Distinct

✓ B. Related

C. Designed

D. All of them

In Access, the structure of a table is created in ___ view.

✓ A. Design View

B. Datasheet View

C. a and b both

D. None of them

Exercise 5c — Question 3 (True / False — book key, p. 241)

An IDE simplifies the tasks of creating and using a database

True

The major objects of database are five

False

Forms are provided by database management system to generate reports

False

An integrated development environment is an interface that is used by database designers and application programmers to create database applications

True

To view data in an Access table, the table is displayed in design view

False

RDBMS stands for Relational Database Management System

True

A request to extract data from a database is called report

False

Database design plays an important role in achieving the goals of efficiency, speed and consistency

True

The table can be displayed in two views in Access. There are Design view and Datasheet view

True

The Window in a database IDE that is used to display, enter and edit data on the screen is called form

True

Exercise 5c — Questions 4-15 (Descriptive)

Define the database objects are used to store and retrieve data.

The four major database objects are tables, queries, forms and reports.

Tables store the data, and queries retrieve specific data from the tables.

Forms display, enter and edit data on the screen, and reports present the retrieved data in a formatted and printed form.

Explain the procedure for creating a new database in Access.

To create a new database in Access, open Microsoft Access and select the option for a blank database.

Give the database a file name and select the location where it will be saved.

Then click the Create button, and the database window opens, in which tables and other objects can be made.

Differentiate between Toolbar and Menu bar.

The menu bar is a row of menus at the top of the Access window, such as File, Edit and View.

The toolbar is a row of buttons with icons for common commands, such as save, cut, copy and paste.

Menus give all the commands, while the toolbar gives quick access to the most used commands.

What is meant by RDBMS?

RDBMS stands for Relational Database Management System.

It is a database management system in which data is stored in tables called relations.

Tables are linked with each other through keys, and Microsoft Access is an example of an RDBMS.

What is an IDE?

IDE stands for Integrated Development Environment.

It is an interface used by database designers and application programmers to create database applications.

An IDE provides tools for creating tables, queries, forms and reports in one window, which simplifies the tasks of creating and using a database.

Define the use of Toolbar in Microsoft Access.

The toolbar in Microsoft Access is used to perform common tasks quickly with buttons.

Buttons are available for opening, saving, printing, cutting, copying and pasting objects.

Using the toolbar saves time, because the user does not have to open menus for every command.

What are the advantages of using a Microsoft Access IDE?

The Microsoft Access IDE simplifies the tasks of creating and using a database.

Tables, queries, forms and reports can be created in one window without writing programs.

It also provides wizards, which guide the user step by step in creating database objects.

Write a procedure to open an existing database file.

To open an existing database file, start Microsoft Access and select the option to open an existing database.

Select the database file from its folder and click the Open button.

The database window appears, showing the tables, queries, forms and reports of the database.

How is Microsoft Access started or loaded?

Microsoft Access is started by clicking the Start button and selecting Microsoft Access from the programs menu.

It can also be started by double clicking the Access icon on the desktop.

After loading, the Access window appears, from which a new database can be created or an existing one opened.

Differentiate between Form and Report.

A form is used to display, enter and edit data on the screen, and it works as the front end of the database.

A report is used only to retrieve data from the database and display it on the screen or print it, and it cannot edit data.

Forms are for data entry, while reports present the finished result of data.

Define the use of status bar and title bar in Microsoft Access.

The title bar is the top bar of the window, which shows the name of the database and the program.

The status bar is the bottom bar, which shows information about the current object

and the current record.

Both bars help the user to know which database is open and what is happening in the window.

Describe the Database Window in Microsoft Access.

The database window is the main window of a database in Microsoft Access.

It shows the objects of the database: tables, queries, forms, reports, pages, macros and modules.

From the database window, any object can be opened, designed, created or deleted.

Chapter 6: Table and Query

Exercise 6c — Question 1 (Fill in the blanks — book key, p. 241)

The Primary key is specified in table to avoid duplicate entries of records.

In Microsoft Access, the data of the table is displayed in Datasheet view.

In the relational model, Table is the basic structure in which data is stored.

The ERD is a graphical representation of the structure of a database.

The Right head arrow button shows the current record in the table.

If the primary key is made up of a group of two or more fields, it is called composite .

A Wildcard are special characters that are used in queries to specify the criteria.

In Microsoft Access, the output of a query is in the form of a Table .

A query that involves two tables is called cross Table .

The wildcard character Asterisk (*) is used to specify any number of characters.

The Memo data type is used when the field is to contain text consisting of about 300 characters.

The number of rows in a table of a relational database is called the Cardinality of the table.

The number of columns in a table is called the Degree of the table.

A query that only retrieve and displays data is called Select .

The wildcard character # is used to specify a single digit.

Exercise 6c — Question 2 (Multiple choice — book key, p. 241)

The data in table is entered in:

A. Design View

B. Normal View

✓ C. Datasheet View

D. Layout View

How many types of relationship?

A. 2

✓ B. 3

C. 4

D. 5

In a relational database, a single piece of information is called:

✓ A. Field

B. Record

C. Entity

D. Attribute

The rule that a record from a table cannot be deleted if its associated record exists in a related table is called ___ rule.

✓ A. Referential integrity

B. Entity - relationship

C. Normalization

D. All of them

How many table views are available in Microsoft Access.

✓ A. 4

B. 3

C. 2

D. 1

To find a four-character name that starts with H, the criteria is specified as.

A. H*4

B. H?4

✓ C. H???

D. H####

Which of the following buttons of Find and Replace dialog box is clicked to start the search process?

A. Find

✓ B. Find Next

C. Search

D. Next

As in Design view, you can move from field to field in the Table window in Datasheet view using ___ button:

✓ A. Tab

B. Esc

C. Enter

D. Spacebar

The relationship between countries and their capitals is an example of ___ relationships.

✓ A. one-to-one

B. one-to-many

C. many-to-many

D. None of them

The wildcard ___ Sal[ei]ma.

A. Saleema

✓ B. Salima

C. both a and b

D. None of them

Exercise 6c — Question 3 (True / False — book key, p. 241)

If for each entity in B, and for each entity in B, there is only one related entity in A, then the relationship between the entities is one-to-many

False

Date/Time data type is used when data such as data of birth or time of day is to be stored

True

The primary key is assigned in a table to avoid duplicate entries of records

True

An append query adds a group of records from one or more tables to the end of one or more tables

True

The currency data type cannot be used in calculations

False

The most commonly used type of query is Scratch Query

False

A query is used to extract specific information from a database

True

The wildcard character '?' is used to specify any number of characters

False

Fields are displayed as check boxes by default on the datasheet

False

An action query makes changes in specified records of an existing table, or creates a new table

True

Exercise 6c — Questions 4, 6-20 (Descriptive; Q5 is skipped in the book's numbering)

Define the different data types available in Microsoft Access.

The data types in Microsoft Access are Text, Memo, Number, Date/Time, Currency, AutoNumber, Yes/No, OLE Object and Hyperlink.

Text stores up to 255 characters, Memo stores long text of about 300 or more characters, and Number stores numeric values.

Date/Time stores dates and times, Currency stores money values, AutoNumber increases automatically, and Yes/No stores true or false values.

Define the primary key.

A primary key is a field, or a group of fields, that uniquely identifies each record in a table.

It is specified in the table to avoid duplicate entries of records.

A primary key value is never null and never repeated, and if it is made of two or more fields it is called a composite key.

What is Referential Integrity?

Referential integrity is the rule that a record from a table cannot be deleted if its associated record exists in a related table.

It also requires that a foreign key value matches a primary key value in another table.

In this way the relationships between tables remain correct and no orphan record is created.

What are relationships?

A relationship is a logical connection between two tables, made through a common field.

There are three types of relationships: one-to-one, one-to-many and many-to-many.

For example, the relationship between countries and their capitals is a one-to-one relationship.

How are relationship defined in Microsoft Access?

In Microsoft Access, relationships are defined in the Relationships window.

The tables are added to the window, and a field of one table is dragged onto the matching field of the other table.

Then the type of relationship is selected, referential integrity is enforced, and the relationship is created.

Explain the options in Find and Replace dialog box.

The Find and Replace dialog box is used to search for a value in a table and replace it with another value.

Its options are Find What, Replace With, Look In, Match, Search and the Find Next button.

Find Next starts the search process, and Replace or Replace All changes the found values.

Differentiate between Relationship and Join.

A relationship is a permanent logical connection defined between two tables in the database.

A join is a temporary connection made in a query to combine records from two or more tables.

Relationships are defined once, while joins are used whenever a query needs data

from more than one table.

Define different types calculation in a query and also specifies the some Functions.

Calculations in a query are made with expressions using arithmetic operators and built-in functions.

Common functions are Sum, Avg, Count, Min and Max, which calculate totals, averages, counts and smallest or largest values.

For example, Sum finds the total of a field, and Count counts the number of records in a query.

What is query? Discuss its uses and advantages.

A query is a database object used to extract specific information from a database.

Its uses are to retrieve selected records, to sort and filter data, and to perform calculations.

The advantages of a query are that only the required data is shown, data from many tables can be combined, and the result is displayed in the form of a table.

Explain the criteria in a query. How is it specified?

Criteria in a query are the conditions that decide which records will be shown in the result.

Criteria are specified in the criteria row of the query design grid under the required field.

For example, to find students with marks above 60, the criteria >60 is written under the marks field, and wildcards such as * and ? can also be used.

What is a join? Explain its purpose.

A join is a connection between two or more tables in a query, made through a common field.

Its purpose is to combine related records from different tables into one result.

For example, a join of the student table and the marks table shows each student's name with his marks.

Differentiate between Sorting and Filtering.

Sorting arranges the records of a table in ascending or descending order of a field.

Filtering displays only those records which satisfy a given condition, hiding the other records.

Sorting changes the order of records, while filtering changes which records are shown.

What are wildcards?

Wildcards are special characters used in queries to specify the criteria.

The asterisk (*) specifies any number of characters, the question mark (?) specifies a single character, and the hash (#) specifies a single digit.

For example, H??? finds a four-character name starting with H.

Define the various types of queries.

The various types of queries are select queries and action queries.

A select query only retrieves and displays data, while an action query makes changes in the data.

Action queries include update queries, delete queries, append queries and make-table queries.

How can you create a query in Design View?

To create a query in Design View, open the query in Design View and add the required tables.

Select the fields to be shown, write the criteria under the fields, and set the sorting order if needed.

Then run the query, and the result is displayed in the form of a table.

Create a Query in Design View.

To create a query in Design View, first select the tables whose data is needed and add them to the design grid.

Then select the fields, write the criteria, and set the sort order for the result.

Finally run the query and save it with a suitable name, so it can be used again.

Chapter 7: Microsoft Access-Forms and Reports

Exercise 7c — Question 1 (Fill in the blanks — book key, p. 241)

Forms are used to Enter, view, modify or delete data in database.

The easiest and quick way to create a form in Microsoft Access is by using Wizard .

A Tabular form is used to display multiple records at a time in tabular format.

A sub-form is a form, which is displayed within the main form.

There are basically Four layouts of forms in Microsoft Access.

The Report object is used only to retrieve data from a database and display it on the screen or print it on the printer. It cannot be used to edit data in a database.

A report are can compare, summarize, and subtotal large sets of data.

In Microsoft Access, a report can be created in any one of the three layouts. These layouts are Columnar, Tabular, Justified .

Reports are the finished result of your data input to the database backed through the forms as front end end.

The Report object is used only to retrieve data from a database and displays it on the screen or print it on the printer. It cannot be used to edit data in a database.

Exercise 7c — Question 2 (Multiple choice — book key, p. 241)

Forms are designed for:

A. Input Data

✓ B. Manipulate Data

C. Accepting Change

D. All of them

A form that contains the sub form is called ___.

A. Form

✓ B. Main Form

C. Report

D. None of them

You can drag the ___ bar to move the property sheet window around on your screen.

✓ A. Title bar

B. Status bar

C. Scroll bar

D. All of them

How many are basic layouts of forms in Microsoft Access.

A. 2

B. 3

✓ C. 4

D. 5

The forms are the ___ end of our database in Microsoft Access.

A. Back end

✓ B. Front end

C. both and b

D. None of them

A ___ Auto form displays one record at a time.

A. Tabular

✓ B. Columnar

C. Datasheet

D. Justified

A report provides a column for each field of the records in rows under the column header is known as:

✓ A. Tabular

B. Columnar

C. Datasheet

D. Justified

___ can be previewed on the screen before printing.

✓ A. Report

B. Form

C. Subform

D. None of them

Which of the following is used to retrieve data from one or more tables of database and to present it to the user in a formatted way?

✓ A. Report

B. Form

C. Query

D. Table

How many are the layout of report?

A. 2

✓ B. 3

C. 4

D. 5

Exercise 7c — Question 3 (True / False — book key, p. 241)

A form can enter, edit and retrieve data from only one table at a time

False

When a form is closed, the records entered in the form are automatically saved in the database

True

We cannot change any record of a table in a database using a form

False

A form within another form is known as tabular form

False

The tabular layout of a form is called datasheet layout

False

We have not any facility to test a conditional statement in reports

False

The easiest way to create a report in Microsoft Access is by using the Report Wizard

True

The options for layout of report are Columnar, Tabular and justified

True

Report are primarily used to input data easily into database tables

False

The record navigation buttons at the lower-left corner of a form are used to save recently edited data in the database

True

Exercise 7c — Questions 4-15 (Descriptive)

Define the different layouts of forms.

The four basic layouts of forms in Microsoft Access are Columnar, Tabular, Datasheet and Justified.

A columnar form displays one record at a time, and a tabular form displays multiple records in rows and columns.

A datasheet form looks like a table, and a justified form displays one record with fields arranged across the width of the form.

Differentiate between forms or reports.

A form is used to enter, view, modify or delete data, and it is the front end of the database.

A report is used only to retrieve data and display it on the screen or print it, and it cannot edit data.

Forms are for data entry, while reports are the finished result of data, used to compare, summarize and subtotal large sets of data.

How is a form created by using the form visit? (Form Wizard)

To create a form with the Form Wizard, select the table and start the Form Wizard from the Forms object.

Select the fields to be shown on the form and choose the layout of the form.

Then give the form a title and finish the wizard, and the form is created and displayed.

Explain the processor for editing data through forms.

To edit data through a form, open the form and move to the record that is to be changed.

Change the values in the fields, and the record is saved automatically when the form is closed or when another record is selected.

In this way data can be entered, viewed, modified or deleted easily through forms.

What is sub form?

A sub-form is a form which is displayed within the main form.

The main form shows a record of one table, and the sub-form shows the related records of another table.

For example, a customer form can contain a sub-form showing all the orders of that customer.

How is a form created for a single table?

To create a form for a single table, select the table in the database window.

Start the Form Wizard or AutoForm, select the fields of the table, and choose the

layout.

Then save the form, and it will display the records of that single table one by one.

What is a report?

A report is a database object used only to retrieve data from a database and display it on the screen or print it on the printer.

Reports cannot be used to edit data in a database.

Reports can compare, summarize and subtotal large sets of data, and they present data in a formatted way.

How can you save and closed an auto report?

An auto report is created with the AutoReport feature, which makes a report from a table at once.

To save it, select the Save option, give the report a name, and click OK.

To close it, click the close button of the report window, and the report is saved in the database.

Define to create a report using the report wizard.

To create a report with the Report Wizard, select the table or query and start the Report Wizard.

Select the fields, group the records if needed, and set the sort order.

Then choose the layout from Columnar, Tabular or Justified, give the report a title, and finish the wizard.

How can you apply a sort order to the report?

To apply a sort order to a report, open the report in Design View and set the sorting fields.

Records are sorted in ascending or descending order of the selected field.

For example, a report of students can be sorted by roll number or by name.

How is a report created from two tables?

To create a report from two tables, first make a query that joins the two tables on a

common field.

Then start the Report Wizard and select the query as the source of the report.

Select the fields from both tables, choose the layout, and the report shows the combined data.

How can report created for a single table?

To create a report for a single table, select the table in the database window.

Start the Report Wizard or AutoReport and select the fields of the table.

Choose the layout, give the report a title, and the report is created showing the records of that table.

Chapter 8: Getting Started with C

Exercise 8c — Question 1 (Fill in the blanks — book key, p. 241)

The set of instruction given to the computer to solve any kind of problem is called Computer program .

ANSI stands for American National Standard Institute .

The program written in high level language is known as Source program .

A Loader is a program that places the executable file in memory.

In Unstructured programming language, the entire logic of the program is implemented in a single module.

A Preprocessor directives are command that give instruction to C preprocessor.

A Constant Macro is a name that is replaced by particular constant before program is sent to the compiler.

The Include directive gives a program access to a library file.

C program is divided into units, called Function .

Every statement in a C program terminates with a Semicolon(;) .

A language translator for Assembly language is called Assembler .

A set of rule for writing program in high level language is known as Syntax .

Exercise 8c — Question 2 (Multiple choice — book key, p. 241)

C is a:

✓ A. High Level Language

B. Low Level Language

C. Assembly Language

D. Machine Language

Turbo C++ can compile:

A. C++ programs only

✓ B. C and C++ programs

C. Turbo C programs only

D. Turbo C++ programs only

Debug is the process of:

A. Creating bugs in program

✓ B. Identifying and removing errors

C. Identifying Errors

D. Removing Errors

C was designed to write programs for:

A. Windows operating system

B. Solaris operating system

✓ C. Unix operating system

D. OS/2 operating system

Preprocessor directives are commands for:

A. Microprocessor

B. Language processor

✓ C. C preprocessor

D. Loader

The expression in define directive:

A. can only be changed at the end of the program

✓ B. can not be changed

C. can not be changed but can be redefined

D. can not be assigned a value

Which of the following language requires no translator to execute the program:

A. C

B. C++

✓ C. Machine language

D. Assembly language

.exe file is produced by the:

✓ A. Linker

B. Loader

C. Compiler

D. Interpreter

Which of the following key is used to save a file?

A. F2

✓ B. F3

C. F5

D. F9

void occupy how many bytes in memory?

✓ A. zero

B. one

C. two

D. four

Exercise 8c — Question 3 (True / False — book key, p. 241)

The C programming language was developed by Dennis Ritchie in 1972

True

C was derived from earlier programming language named B

True

The B was developed by Ken Thomson in 1980

False

The short-key for compiling a program is Alt+F9

True

The compiler produces the source file from an object file

False

The linker is a program that combines the object program with additional files

True

The short-key for executing the C program is Alt + F5

False

In structured programming the entire program is divided into smaller modules

True

The value of a constant can be changed during the program execution

False

High level language provide machine independence

True

Exercise 8c — Questions 4-13 (Descriptive)

Briefly describe the history of C.

C was developed by Dennis Ritchie in 1972 at Bell Laboratories.

It was derived from the earlier programming language B, which was developed by Ken Thompson.

C was designed to write programs for the Unix operating system, and later it became a popular general purpose language.

List two reasons why it would be preferable to write a program in C rather than machine language.

Programs in C are easy to write and understand, because C uses English-like words instead of binary codes.

C provides machine independence, so a C program written on one computer can be run on another computer after compiling.

Machine language programs are difficult to write and can run only on one type of machine.

What necessary steps taken to prepare a C program for execution? Explain with diagram.

To prepare a C program for execution, first the source program is written in an editor.

Then the program is compiled into an object program, and the linker combines the object program with library files to make an executable (.exe) file.

Finally the loader places the executable file in memory and the program is executed.

Define a bug. Discuss some debugging features of Turbo C++.

A bug is an error in a program which stops it from giving correct results.

Debugging is the process of identifying and removing errors from a program.

Turbo C++ provides debugging features such as step by step execution, watching the values of variables, and setting break points in the program.

While writing a C program, how many types of errors can occur? Discuss briefly. Which one is the most difficult to locate and remove? Justify your answer.

While writing a C program, three types of errors can occur: syntax errors, run time errors and logical errors.

Syntax errors break the rules of the language and are found by the compiler, and run time errors occur during execution.

Logical errors are the most difficult to locate and remove, because the program runs but gives wrong results, and the compiler gives no message about them.

What is a programming language? Discuss the two main categories of programming languages.

A programming language is a set of rules for writing programs, which are sets of instructions given to the computer to solve a problem.

The two main categories are low level languages and high level languages.

Low level languages are machine language and assembly language, while high level languages are C, C++ and Java, which are easy to understand.

Describe characteristics of high-level programming languages.

High level programming languages use English-like words, so they are easy to learn and understand.

They provide machine independence, so the same program can run on different computers.

They also provide structured programming, in which a program is divided into smaller modules, and one statement can perform many machine operations.

Briefly describe the basic structure of a C program.

The basic structure of a C program starts with preprocessor directives, such as #include <stdio.h>.

Then global declarations are written, followed by the main() function, which contains variable declarations and statements.

Every statement ends with a semicolon, and user-defined functions are written after the main function.

How would you create, edit, compile, link and execute a C program? Discuss briefly.

To create a C program, open the editor of Turbo C++ and write the source program,

then save the file with the F3 key.

To edit it, open the file again and change the program as needed.

To compile it press Alt+F9, the linker makes the .exe file, and the program is executed with Ctrl+F9.

Differentiate the following: (i) Preprocessor Directive and the Compiler (ii) Structured and Unstructured programming languages (iii) Linker and Loader.

A preprocessor directive gives instructions to the C preprocessor before compilation, while the compiler translates the source program into an object program.

Structured programming divides a program into smaller modules, while unstructured programming implements the entire logic in a single module.

The linker combines the object program with library files to make an executable file, while the loader places the executable file in memory.

Chapter 9: Elements of C

Exercise 9c — Question 1 (Fill in the blanks — book key, p. 241)

The first character of a variable name must be a Letter .

A Binary operator operates on two operands.

Named memory cells which are used to store program's input and output are called Variable .

The maximum length of a character constant is One character.

The value of a variable of type int ranges from -32768, 32767 .

The value of an unsigned int variable ranges from 0 to 65535 .

The value of a float variable ranges from 10^-38, 10^38 .

The symbol for logical OR operator is || .

A Comments are used to increase the readability of the program.

Multi-line comments start with /* and end with */.

Exercise 9c — Question 2 (Multiple choice — book key, p. 241)

Variables are created in:

✓ A. RAM

B. ROM

C. Hard Disk

D. Cache

Which of the following is a valid character constant?

A. a

B. "b"

✓ C. '6'

D. =

Which of the following data type offers the highest precision?

A. float

B. long int

✓ C. long double

D. unsigned long int

When the result of the computation of two very small numbers is too small to be represented, this phenomenon is called:

A. Arithmetic overflows

✓ B. Arithmetic underflow

C. Truncation

D. Round off

The symbol '=', represents:

A. Comparison operator

✓ B. Assignment operator

C. Equal-to operator

D. None of these

Which of the following operators has lowest precedence?

A. !

B. +

✓ C. =

D. ==

Relational operators are used to:

A. Establish a relationship among variables

✓ B. Compare two values

C. Construct compound condition

D. Perform arithmetic operations

C is a strongly typed language, this means that:

A. Every program must be compiled before execution

✓ B. Every variable must be declared before it is being used

C. The variable declaration also defines the variable

D. Sufficient data types are available to manipulate each type of data

The logical not operator, denoted by !, is a:

A. Ternary operator

✓ B. Uniary operator

C. Binary operator

D. Bitwise operator

a += b is equivalent to:

A. b += a

B. a =+ b

✓ C. a = a + b

D. b = b + a

Exercise 9c — Question 3 (True / False — book key, p. 241)

printf and scanf are standard identifiers

True

In C language, all variables must be declared before being used

True

Standard data types are not predefined in C language

False

The double data type required 4 bytes in memory

False

In Scientific notation, the exponent represents the value of the number and mantissa represents the power to which it is raised

False

The symbol for modulus operator is %

True

The symbol = is used to compare two values

False

Operator precedence determines the order of evaluation of operators in an expression

True

For many compilers a C variable name can be up to 31 characters

True

C program can only use lowercase letters in variable names

False

Exercise 9c — Questions 4-14 (Descriptive)

What is an identifier? Discuss the two types of identifiers in C.

An identifier is a name given to a variable, function or other item in a C program.

The two types of identifiers are standard identifiers and user-defined identifiers.

Standard identifiers, such as printf and scanf, are provided by the language, while user-defined identifiers are made by the programmer.

What is a variable? Discuss the difference between declaring and defining a variable.

A variable is a named memory cell used to store a program's input and output.

Declaring a variable tells the compiler its name and type, while defining a variable also allocates memory for it.

In C, a declaration like int a; both declares and defines the variable.

Write down rules for naming variables in C.

The first character of a variable name must be a letter or an underscore.

The remaining characters can be letters, digits or underscores, and the name cannot be a keyword.

For many compilers a variable name can be up to 31 characters, and uppercase and lowercase letters are treated as different.

Differentiate the following: (i) Constant and variable (ii) Character constant and Numeric constant (iii) Standard data type and User defined data type (iv) Keyword and Identifier.

A constant is a value that cannot change during execution, while a variable can change.

A character constant is a single character in single quotes, such as 'A', while a numeric constant is a number, such as 25.

Standard data types like int and float are predefined, user-defined types are made by the programmer, keywords are reserved words, and identifiers are names given by the programmer.

What is a data type? Discuss various C data types to manipulate integers, floating point numbers and characters.

A data type tells the compiler what kind of value a variable will hold and how much memory it needs.

For integers, int (range -32768 to 32767) and long int are used; for floating point numbers, float and double are used.

For characters, char is used, which stores one character such as 'A' or '6'.

How many types of operators are available in C? Describe briefly. Also mention their precedence.

C provides arithmetic operators (+, -, *, /, %), relational operators (<, >, <=, >=, ==, !=), logical operators (&&, ||, !) and assignment operators (=, +=, -=).

It also provides unary operators, such as increment and decrement, and the conditional operator.

In precedence, unary operators are highest, then arithmetic, then relational, then logical, and assignment operators are lowest.

What data type would you use to represent the following items: number of children at your school, a letter grade on an exam, and the average marks of your class?

The number of children at school is a whole number, so the int data type is used.

A letter grade on an exam is a single character, so the char data type is used.

The average marks of the class can have a fractional part, so the float data type is used.

Which of the following are valid variable names in C? (i) income (ii) total marks (iii) double (iv) average-score (v) room# (vi) _area (vii) no_of_students (viii) long (ix) Item (x) MAX_SPEED.

Valid names are income, _area, no_of_students, Item and MAX_SPEED.

Invalid names are total marks (space is not allowed), double and long (both are keywords), average-score (hyphen is not allowed) and room# (# is not allowed).

The first character must be a letter or underscore, and keywords can never be used as variable names.

Write a note on the following: (i) Arithmetic Expression (ii) Comments in C.

An arithmetic expression is a combination of variables, constants and arithmetic operators that gives a numeric value, such as a + b * c.

Comments in C are used to increase the readability of the program and are ignored by the compiler.

Single-line comments start with //, and multi-line comments start with /* and end with */.

Let w, x, y, and z be the name of four type float variables, and let a, b and c be the names of three type int variables. Each of the following statements contains one or more violations of the rules for forming arithmetic expressions. Rewrite these statements so that it is consistent with these rules. (i) z = 4.0 w * y; (ii) y = yz; (iii) a = 6b4; (iv) c = 3(a + b); (v) z = 7w + xy;

The corrected statements are: (i) z = 4.0 * w * y; (ii) y = y * z; (iii) a = 6 * b * 4; (iv) c = 3 * (a + b); (v) z = 7 * w + x * y.

In C, multiplication must always be written with the * operator, and two names or numbers cannot be written side by side.

A number before a bracket must also be joined with *, as in 3 * (a + b).

Assume that you have the following variable declarations: int a, b, c, d, p; float v, w, x, y, z; Evaluate each of the following statements assuming a is 2, z is -1.3, c is 1, d is 3, y is 0.3E+1. (i) v = a * 2.5 / y; (ii) w = a / y; (iii) p = a / d; (iv) x = (a + c) / (z + 0.3); (v) b = d / a + d % a; (vi) y = c / d * a;

Given a = 2, z = -1.3, c = 1, d = 3 and y = 0.3E+1 = 3.0, the results are:

(i) v = 2 * 2.5 / 3.0 = 1.666667. (ii) w = 2 / 3.0 = 0.666667. (iii) p = 2 / 3 = 0, because integer division drops the fraction.

(iv) x = (2 + 1) / (-1.3 + 0.3) = 3 / -1.0 = -3.0. (v) b = 3 / 2 + 3 % 2 = 1 + 1 = 2. (vi) y = 1 / 3 * 2 = 0 * 2 = 0, because 1 / 3 is integer division.

All six results were verified by compiling and running the expressions with gcc.

Chapter 10: Input / Output

Exercise 10c — Question 1 (Fill in the blanks — book key, p. 242)

The getch function does not display characters on the output screen.

A scanf is an input function.

%x is a format specifier for Hexadecimal integers .

Escape sequences always begins with a Backslash (\) .

The printf function is defined in stdio.h .

The ASCII code for Escape key is 27 .

The escape sequence \r represents the carriage return.

There are total 80 columns on the output screen.

The symbol for address of operator is & .

\ddd is used to print ASCII code in Octal notation.

Exercise 10c — Question 2 (Multiple choice — book key, p. 242, with flagged items)

The function getche() is defined in:

A. stdio.h

B. string.h

C. math.h

✓ D. conio.h

The escape sequence for backslash is:

A. \

B. \b

✓ C. \\

D. \t

The format specifier %u is used for:

A. integer

✓ B. unsigned short

C. unsigned float

D. unsigned long int

When the result of the computation of two very small numbers is too small to be represented, this phenomenon is called:

A. Arithmetic overflows

✓ B. Arithmetic underflow

C. Truncation

D. Round off

The symbol '=', represents:

A. Comparison operator

✓ B. Assignment operator

C. Equal-to operator

D. None of these

Which of the following operators has lowest precedence?

A. !

B. +

✓ C. =

D. ==

Relational operators are used to:

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✓ B. Compare two values

C. Construct compound condition

D. Perform arithmetic operations

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C. The variable declaration also defines the variable

D. Sufficient data types are available to manipulate each type of data

The logical not operator, denoted by !, is a:

A. Ternary operator

✓ B. Uniary operator

C. Binary operator

D. Bitwise operator

a += b is equivalent to:

A. b += a

B. a =+ b

✓ C. a = a + b

D. b = b + a

Exercise 10c — Question 3 (True / False — the four items printed in the book, p. 173)

printf and scanf are standard identifiers

True

In C language you must declare all variables before using them

True

Standard data types are not predefined in C language

False

The double data type required 4 bytes in memory

False

Exercise 10c — Questions 4-12 (Descriptive and programs)

What do we mean by standard input and output? Illustrate the use of printf() and scanf() functions.

Standard input means data entered from the keyboard, and standard output means data displayed on the screen.

The printf() function is used for standard output, for example printf("%d", a); displays the value of a.

The scanf() function is used for standard input, for example scanf("%d", &a); reads a number from the keyboard into a.

Illustrate the difference between format specifiers and field-width specifiers with examples.

A format specifier tells the type of data, such as %d for integers, %f for float and %c for characters.

A field-width specifier tells how many columns the value will occupy, such as %5d, which prints an integer in 5 columns.

For example, printf("%5d", 25); prints three spaces followed by 25, while printf("%d", 25); prints only 25.

Define the term 'escape sequence'. List names and uses of any five escape sequences.

An escape sequence is a combination of a backslash and a character, used to print special characters or control the output.

Five escape sequences are: \n (new line), \t (horizontal tab), \\ (backslash), \' (single quote) and \" (double quote).

Escape sequences always begin with a backslash, and \r represents the carriage return.

Show the output displayed by the following program when the data entered are 10 and 15. (a) Output (b) Memory contents (c) Rewrite using scanf.

Note: the program as printed in the book writes scanf("%d %d", m, n); without the &

signs; the corrected form scanf("%d %d", &m, &n); is used below.

(a) Output: the program reads m = 10 and n = 15, then m = 10 + 10 = 20 and n = 5 * 20 = 100.

The output is: m = 20 (followed by three tabs) n = 100, verified by compiling and running the program with gcc.

(b) Memory contents: before execution m and n are uninitialized; after execution m holds 20 and n holds 100.

(c) The program in example 5 rewritten with scanf reads its input values with scanf instead of fixed assignment statements, for example scanf("%d %d", &a, &b);

Show how the value -17.246 would be printed using the formats %8.4f, %8.3f, %8.2f, %8.1f, %8.0f, and %0.2f.

The outputs, verified with gcc, are:

%8.4f prints -17.2460. %8.3f prints one space followed by -17.246. %8.2f prints two spaces followed by -17.25.

%8.1f prints three spaces followed by -17.2. %8.0f prints five spaces followed by -17. %0.2f prints -17.25 with no padding.

Assuming x (type double) is 21.335 and y (type int) is 200, show the output of the following statements (on paper). For clarity, use the symbol box to denote a blank space. printf("x is %6.2f \t y is %4d\n", x, y); printf("y is %d\n", y); printf("x is %.1f\n", x);

The outputs, verified with gcc, are:

First statement: x is, one space, 21.34, one space, tab, y is, one space, 200.

Second statement: y is 200. Third statement: x is 21.3.

If the variables a, b, and c are 307, 408.558 and -12.31, respectively, write a statement that will display the following line: (box shows a blank space) two boxes, 307, five boxes, 408.56, three boxes, -12.3.

The statement is: printf("%5d%11.2f%8.1f", a, b, c);

%5d prints 307 in 5 columns (two spaces before 307), %11.2f prints 408.56 in 11 columns (five spaces before it), and %8.1f prints -12.3 in 8 columns (three spaces before it).

The result, verified with gcc, is: two spaces, 307, five spaces, 408.56, three spaces, -12.3.

Write a program that asks the user to enter the radius of a circle and then computes and displays the circle's area. Use the formula area = PI x radius x radius, where PI is the constant value of 3.14159. (Note: Define a constant macro PI with #define directive).

```
#include <stdio.h>
#define PI 3.14159
int main(void)
{
float radius, area;
printf("Enter radius: ");
scanf("%f", &radius);
area = PI * radius * radius;
printf("Area = %f\n", area);
return 0;
}
```

Write a program that stores the values 'A', 'U', 3.456E10 and 50 in separate memory cells. Your program should get the first three values as input data, but use an assignment statement to store the last value.

```
#include <stdio.h>
int main(void)
{
char c1, c2;
```

```
double d;
int n;
printf("Enter a character, a character and a number: ");
scanf(" %c %c %lf", &c1, &c2, &d);
n = 50;
printf("%c %c %e %d\n", c1, c2, d, n);
return 0;
}
```

The first three values ('A', 'U', 3.456E10) are read with scanf, and the last value 50 is stored with the assignment statement n = 50.

Write a program that converts a temperature in degrees Fahrenheit to degrees Celsius. For conversion, use the following formula: celsius = 5/9 (fahrenheit - 32).

```
#include <stdio.h>
int main(void)
{
float fahrenheit, celsius;
printf("Enter temperature in Fahrenheit: ");
scanf("%f", &fahrenheit);
celsius = 5.0 / 9.0 * (fahrenheit - 32);
printf("Celsius = %f\n", celsius);
return 0;
}
```

Note: 5.0 / 9.0 is written instead of 5 / 9, because in C the integer division 5 / 9 gives 0.

Write a program that takes a positive number with a fractional part and rounds it to two decimal places. For example, 25.4851 would round to 25.49, and 62.4431 would round to 62.44.

```
#include <stdio.h>
int main(void)
{
float num;
printf("Enter a number: ");
scanf("%f", &num);
printf("Rounded value = %.2f\n", num);
return 0;
}
```

The %.2f format rounds the value to two decimal places, so 25.4851 prints as 25.49 and 62.4431 prints as 62.44.

(Note: the book prints '32.44' for the second example, which is a printing error; the correct rounded value of 62.4431 is 62.44.)

Chapter 11: Decision Constructs

Exercise 11c — Question 1 (Fill in the blanks — book key, p. 242)

A control structure control the flow of execution in a program.

A compound statement is a group statements enclosed in braces.

if statement is used to select a pathflow in a program based on a condition .

A flowchart is a pictorial representation of a program.

A nested if refers to a structure with one if statement inside another if statement.

A switch-case statement can be used as an alternative to if-else if statement.

In switch statement the value of expression must be integer, character or character.

The break statement switches the control outside the block in which it is used.

The purpose of default label in switch-case statement is the same as that of else in if-else statement.

A condition is an expression that is either true, false or false.

Exercise 11c — Question 2 (True / False — book key, p. 242)

An arithmetic expression can not be used as condition in if statement

True

The conditional operator is a ternary operator

True

Logical operators are used to compare values

False

A switch statement can not be used within the block of an if statement

False

The break statement stops the execution of a program for a moment and then resumes

True

In sequence structure, statements are executed in the same order in which they appear in the program

True

A false condition always evaluates to zero

False

Use of the statement terminator at the end of an if statement causes a syntax error

False

The switch expression can not be of float or double type

True

C is an unstructured programming language

False

Exercise 11c — Questions 3-12 (Descriptive and programs)

What is a control structure? Briefly describe the basic control structures for writing programs.

A control structure controls the flow of execution in a program.

The basic control structures are sequence, selection and repetition.

In sequence, statements run in order; in selection, one path is chosen by a condition; in repetition, statements run again and again while a condition is true.

How many selection statements are available in C? Discuss the difference between them.

The selection statements available in C are if, if-else, nested if, if-else-if ladder, switch and the conditional operator.

The if statement selects one path, if-else selects between two paths, and the if-else-if ladder selects among many paths.

The switch statement selects one case from many constant values, and the conditional operator selects between two values in a single expression.

Write the general form of the following statements: (i) if statement with one alternative (ii) if statement with two alternatives (iii) if statement with multiple alternatives (iv) switch statement.

The general forms are:

(i) if (condition) statement; (ii) if (condition) statement1; else statement2;

(iii) if (condition1) statement1; else if (condition2) statement2; else statement3;

(iv) switch (expression) { case value1: statements; break; case value2: statements; break; default: statements; }

Rewrite the program given in the example 4 using if statement.

Note: example 4 is a program in the chapter text (a switch statement program); its if-else equivalent is given here.

A switch such as switch (choice) { case 1: ...; break; case 2: ...; break; default: ...; }

is rewritten as:

if (choice == 1) { ...; } else if (choice == 2) { ...; } else { ...; }

In this way every case of the switch becomes one condition of the if-else-if ladder, and the default case becomes the final else.

Assuming x is 10.0 and y is 15.0, what are the values of the following conditions: (a) x != y (b) x < x (c) x >= y - x (d) x == y + x - y. [Hint: The value of a true condition is 1 and a false condition is 0].

With x = 10.0 and y = 15.0, the values are:

(a) x != y is 1, because 10.0 is not equal to 15.0. (b) x < x is 0, because 10.0 is not less than itself.

(c) x >= y - x is 1, because 10.0 >= 15.0 - 10.0 = 5.0. (d) x == y + x - y is 1, because y + x - y = 10.0, which equals x.

These results were verified by compiling and running the expressions with gcc.

Write an expression to test each of the following relationships. (a) age is from 18 to 25. (b) temperature is less than 40.0 and greater than 25.0. (c) year is divisible by 4 (Hint: use %). (d) speed is not greater than 80. (e) y is greater than x and less than z. (f) w is either equal to 6 or not greater than 3.

The expressions are:

(a) age >= 18 && age <= 25. (b) temperature < 40.0 && temperature > 25.0. (c) year % 4 == 0.

(d) speed <= 80. (e) y > x && y < z. (f) w == 6 || w <= 3.

Write assignment statement for the following: (a) Assigns a value of 1 (one) to the variable test if k is in the range -m through +m, inclusive. Otherwise, assigns a value of zero. (b) Assigns a value of one (1) to the variable lowercase if ch is a lowercase letter; otherwise, assigns a value of zero (0). (c) Assigns a value of one (1) to the variable divisor if m is a divisor of n; otherwise, assigns a value of zero (0).

The assignment statements, using the conditional operator, are:

(a) test = (k >= -m && k <= m) ? 1 : 0;

(b) lowercase = (ch >= 'a' && ch <= 'z') ? 1 : 0;

(c) divisor = (n % m == 0) ? 1 : 0;

Write an interactive program that contains an if statement that may be used to compute the area of a square (area = side^2) or a triangle (area = 1/2 x base x height) after prompting the user to type the first character of the figure name (S or T).

```
#include <stdio.h>
int main(void)
{
char ch;
float side, base, height, area;
printf("Enter S for square or T for triangle: ");
scanf(" %c", &ch);
if (ch == 'S' || ch == 's')
{
printf("Enter side: ");
scanf("%f", &side);
area = side * side;
printf("Area of square = %f\n", area);
}
else if (ch == 'T' || ch == 't')
{
printf("Enter base and height: ");
scanf("%f %f", &base, &height);
area = 0.5 * base * height;
```

```
printf("Area of triangle = %f\n", area);
}
return 0;
}
```

A year is a leap year if it is divisible by four, except that any year divisible by 100 is a leap year only if it is divisible by 400. Write a program that inputs a year such as 1996, 1800, and 2010, and displays 'Leap year' if it is a leap year, otherwise displays 'Not a leap year'.

```
#include <stdio.h>
int main(void)
{
int year;
printf("Enter year: ");
scanf("%d", &year);
if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
printf("Leap year\n");
else
printf("Not a leap year\n");
return 0;
}
```

For example, 1996 is a leap year, while 1800 and 2010 are not leap years.

Write a program that inputs temperature and displays a message according to the following data: Greater than 35: Hot day. Between 25 and 35 (inclusive): Pleasant day. Less than 25: Cool day.

```
#include <stdio.h>
int main(void)
{
```

```
float temp;
printf("Enter temperature: ");
scanf("%f", &temp);
if (temp > 35)
printf("Hot day\n");
else if (temp >= 25)
printf("Pleasant day\n");
else
printf("Cool day\n");
return 0;
}
```

Write a program that inputs obtained marks of a student, calculates percentage (assume total marks are 1100), and displays his/her grade. The grade should be calculated according to the following rules: >= 80: A+, 70 (inclusive) to 80: A, 60 (inclusive) to 70: B, 50 (inclusive) to 60: C, 40 (inclusive) to 50: D, 33 (inclusive) to 40: E, Less than 33: F.

```
#include <stdio.h>
int main(void)
{
float marks, per;
printf("Enter obtained marks (out of 1100): ");
scanf("%f", &marks);
per = marks / 1100 * 100;
if (per >= 80) printf("Grade: A+\n");
else if (per >= 70) printf("Grade: A\n");
else if (per >= 60) printf("Grade: B\n");
else if (per >= 50) printf("Grade: C\n");
```

```
else if (per >= 40) printf("Grade: D\n");
else if (per >= 33) printf("Grade: E\n");
else printf("Grade: F\n");
return 0;
}
```

Write a program that inputs two numbers and asks for the choice of the user, if user enters 1 then displays the sum of numbers, if user enters 2 then displays result of subtraction of the numbers, if user enters 3 then displays the result of multiplication of the numbers and if user enters 4 then displays result of the division of the numbers (divide the larger number by the smaller number), otherwise displays the message 'Wrong Choice'. [Use a switch statement to implement the solution].

```
#include <stdio.h>
int main(void)
{
float a, b;
int choice;
printf("Enter two numbers: ");
scanf("%f %f", &a, &b);
printf("Enter choice (1-4): ");
scanf("%d", &choice);
switch (choice)
{
case 1: printf("Sum = %f\n", a + b); break;
case 2: printf("Difference = %f\n", a - b); break;
case 3: printf("Product = %f\n", a * b); break;
case 4:
if (a > b) printf("Division = %f\n", a / b);
```

```
else printf("Division = %f\n", b / a);
break;
default: printf("Wrong Choice\n");
}
return 0;
}
```

Chapter 12: Loop Constructs

Exercise 12c — Question 1 (Fill in the blanks — book key, p. 242)

There are three types of loop in C.

The loop condition controls the loop repetition .

In do-while loop, first the body of the loop is executed and then the test condition is checked.

A nested loop means a loop within the body of another loop.

The goto statement performs an unconditional transfer of control to the named label.

Repetition of statements in a program is called loop .

There are three expressions in for loop statement.

The body of the while loop executes only if the specified condition is true .

Increase in the level of nesting increases the complexity of the nested loop.

A label is meaningful only to a goto statement.

Exercise 12c — Question 2 (True / False — book key, p. 242)

There is no difference between while and do-while loop

False

The body of a while loop may or may not execute

True

The do-while loop always executes at least once

True

var++ is an example of prefix increment

False

The condition of an infinite loop never becomes true

False

Initialization expression is optional in for loop

True

for(i = 1; i <= 10; i++); is an infinite loop

False

Loop is a decision making construct

False

A while loop can not be used in the body of a for loop

False

In type casting, a variable of one type behaves as the variable of another type temporarily

True

Exercise 12c — Questions 3-7, 9-14 (Descriptive and programs; Q8 is skipped in the book's numbering)

Define a loop. How many loops are available in C? Compare the following loops: a) while loop and do-while loop b) while loop and for loop.

A loop is a control structure that repeats a group of statements while a condition is true.

Three loops are available in C: while, do-while and for.

In a while loop the condition is checked first, so the body may never run; in a do-while loop the body runs first, so it runs at least once.

A for loop is used when the number of repetitions is known, because its initialization, condition and increment are written together, while a while loop is used when the number of repetitions is not known in advance.

What is a sentinel controlled loop and how it is implemented? Discuss some of the situations where it can be useful.

A sentinel controlled loop is a loop that continues until a special sentinel value is entered by the user.

It is implemented by reading a value inside the loop and testing whether it equals the sentinel value.

It is useful when the number of inputs is not known in advance, such as reading marks until -1 is entered, or reading numbers until the user enters 0.

Write the output of the following program fragments: (a) k = 0; while (k <= 5) { printf("%3d %3d\n", k, 10 - k); k++; } (b) Trace the output of the following piece of code. j = 10; for (int i = 1; i <= 5; ++i) { printf("%d %d\n", i, j); j -= 2; }

The outputs, verified with gcc, are:

(a) Each row prints k and 10 - k in fields of width 3:

0 10, 1 9, 2 8, 3 7, 4 6, 5 5.

(b) The loop prints i and j, and j decreases by 2 each time:

1 10, 2 8, 3 6, 4 4, 5 2.

Correct the following code segments according to the given instructions: (a) Insert braces where they are needed and correct errors if any. The corrected code should accept five integers and should display their sum. (Given code has: count = 0; while (count <= 5); count += 1; printf("Next Number >"); scanf("%d", &next_num); next_num += sum; printf("%d numbers were added; \n", count); printf("Their sum is %d.\n", sum);)

The errors are: a semicolon after while (count <= 5), which makes an empty loop; the condition should be count < 5 for five numbers; sum is never initialized; and next_num += sum is reversed.

The corrected code is:

```
int count = 0, next_num, sum = 0;
while (count < 5)
{
count += 1;
printf("Next Number >");
scanf("%d", &next_num);
sum += next_num;
}
printf("%d numbers were added;\n", count);
printf("Their sum is %d.\n", sum);
```

(b) Rewrite the following code segment using a do-while statement: sum = 0; for (odd = 1; odd < n; odd = odd + 2) sum = sum + odd; printf("Sum of the positive odd numbers less than %d is %d\n", n, sum);

The do-while version is:

```
sum = 0;
odd = 1;
do
{
sum = sum + odd;
odd = odd + 2;
} while (odd < n);
printf("Sum of the positive odd numbers less than %d is %d\n", n, sum);
```

The body runs first and then the condition odd < n is tested, so the same odd

numbers are added.

Trace the output of the following program segments, assuming m is 3 and n is 5: (a) for (k = 1; k <= n; ++k) { for (j = 0; j < k; ++j) { printf("*"); } printf("\n"); } (b) for (k = n; k > 0; --k) { for (j = m; j > 0; --j) { printf("*"); } printf("\n"); }

The outputs, verified with gcc, are:

(a) The inner loop prints k stars for k = 1 to 5, giving the triangle:

*, **, ***, ****, *****.

(b) The outer loop runs 5 times and the inner loop always prints 3 stars, giving five rows of ***.

(c) (a) Re-write the program in example 3 by replacing the inner while loop with a for loop and a do-while loop. (b) Re-write the program in example 3 by replacing the outer for loop with a while loop and a do-while loop.

Note: example 3 is a nested-loop program in the chapter text; the conversion pattern is shown with the star-triangle program.

(a) Inner while loop replaced by for: for (j = 0; j < k; ++j) printf("*");

Inner while loop replaced by do-while: j = 0; do { printf("*"); j++; } while (j < k);

(b) Outer for loop replaced by while: k = 1; while (k <= n) { inner loop; k++; }

Outer for loop replaced by do-while: k = 1; do { inner loop; k++; } while (k <= n);

Write a program that inputs a number and displays the message 'Prime number' if it is a prime number, otherwise displays 'Not a prime number'.

```
#include <stdio.h>
int main(void)
{
int n, i, prime = 1;
printf("Enter a number: ");
scanf("%d", &n);
if (n < 2) prime = 0;
```

```
for (i = 2; i * i <= n; i++)
{
if (n % i == 0) { prime = 0; break; }
}
if (prime) printf("Prime number\n");
else printf("Not a prime number\n");
return 0;
}
```

Write a program that displays the first 15 even numbers.

```
#include <stdio.h>
int main(void)
{
int i, count = 0, num = 2;
while (count < 15)
{
printf("%d ", num);
num += 2;
count++;
}
printf("\n");
return 0;
}
```

The program prints 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30.

Write a program that inputs a number, and displays its table according to the following format: Suppose the number entered is 5, the output will be: 5 * 1 = 5, 5 * 2 = 10, 5 * 3 = 15, ..., 5 * 10 = 50.

```
#include <stdio.h>
```

```
int main(void)
{
int n, i;
printf("Enter a number: ");
scanf("%d", &n);
for (i = 1; i <= 10; i++)
printf("%d * %d = %d\n", n, i, n * i);
return 0;
}
```

Write a program using do-while loop that repeatedly prompts for and takes input until a value in the range 0 through 15 inclusive is input. The program should add all the values before exiting the loop and displays their sum at the end.

```
#include <stdio.h>
int main(void)
{
int num, sum = 0;
do
{
printf("Enter a number: ");
scanf("%d", &num);
sum += num;
} while (num < 0 || num > 15);
printf("Sum = %d\n", sum);
return 0;
}
```

The loop keeps asking for a number until a value from 0 to 15 inclusive is entered, and the sum of all values entered is displayed.

Write a program that produces the following output: 0 / 0 1 / 0 1 2 / 0 1 2 3 / 0 1 2 3 4 / 0 1 2 3 4 5.

```
#include <stdio.h>
int main(void)
{
int i, j;
for (i = 0; i <= 5; i++)
{
for (j = 0; j <= i; j++)
printf("%d ", j);
printf("\n");
}
return 0;
}
```

The output, verified with gcc, is the triangle 0, then 0 1, then 0 1 2, up to 0 1 2 3 4 5.

Write a program the produces the following output: 0 1 / 1 2 / 2 4 / 3 8 / 4 16 / 5 32 / 6 64.

```
#include <stdio.h>
int main(void)
{
int i, p = 1;
for (i = 0; i <= 6; i++)
{
printf("%d %d\n", i, p);
p *= 2;
}
```

```
return 0;
}
```

The output, verified with gcc, is 0 1, 1 2, 2 4, 3 8, 4 16, 5 32, 6 64.

Chapter 13: Functions in C

Exercise 13c — Question 1 (Fill in the blanks — book key, p. 242)

A function is a self contained piece of code.

Pre-defined functions are packaged in libraries .

A prototype provide basic information about the function to the compiler.

The duration for which a variable exists in memory is called its lifetime .

A scope of a variable refers to the region of the program where it can be referenced.

A global variables are declared outside all blocks.

A function can not return more than one value(s) through return statement.

The parameters specified in the function header are called formal parameter.

The parameters passed to a function in the function call are called actual parameters.

Functions help to achieve modular programming.

Exercise 13c — Question 2 (Multiple choice — book key, p. 242)

Function prototypes for built-in functions are specified in:

A. source files

✓ B. header files

C. object files

D. image files

Global variables are created in:

✓ A. RAM

B. ROM

C. hard disk

D. cache

Which of the following is true about a function call?

A. Stops the execution of the program

✓ B. Transfers control to the called function

C. Transfers control to the main function

D. Resumes the execution of the program

Which of the following looks for the prototypes of functions used in a program?

A. linker

B. loader

✓ C. compiler

D. parser

Memory is allocated to a local variable at the time of its:

A. declaration

B. destruction

✓ C. definition

D. first reference

The name of actual and formal parameters:

✓ A. may or may not be same

B. must be same

C. must be different

D. must be in lowercase

Formal arguments are also called:

A. actual arguments

✓ B. dummy arguments

C. original arguments

D. referenced arguments

printf() is a:

✓ A. built-in function

B. user-defined function

C. local function

D. keyword

A built-in function:

A. can not be redefined

✓ B. can be redefined

C. can not return a value

D. should be redefined

In a C program, two functions can have:

A. same name

B. same parameters

C. same name and same parameters

✓ D. same name but different parameters

Exercise 13c — Question 3 (True / False — book key, p. 242)

In C, arguments can be passed to a function only by value

False

There can be multiple main functions in a C program

False

A function can be called anywhere in the program

True

In C, every function must return a value

False

A user-defined function can not be called in another user-defined function

False

A function can be called only once in a program

False

Scope of a local variable is the block in which it is defined

True

Global variables exist in memory till the execution of the program

True

An unstructured program is more difficult to debug than a structured program

True

Function body is an optional part of the function

False

Exercise 13c — Questions 4-15 (Descriptive and programs)

What is a function? How many types of functions are used in C? Discuss the difference between them.

A function is a self contained piece of code that performs one task.

Two types of functions are used in C: built-in (library) functions and user-defined functions.

Built-in functions, such as printf(), are provided in libraries, while user-defined functions are written by the programmer for a specific task.

Differentiate the following: (i) Function Definition and Function Declaration (ii) Global and Local variables (iii) Scope and Lifetime of a variable (iv) Function prototype and Function header (v) Formal parameters and Actual parameters of a function.

A function declaration (prototype) gives basic information about the function to the

compiler, while the definition contains the actual body of the function.

Global variables are declared outside all blocks and exist during the whole program, while local variables exist only inside their block.

Scope is the region where a variable can be referenced, and lifetime is the duration for which it exists in memory.

A prototype ends with a semicolon, a header does not, and formal parameters are in the header while actual parameters are passed in the call.

How is a function call made in a C program? Discuss briefly.

A function call is made by writing the function name followed by the actual parameters in brackets.

The call transfers control to the called function, which runs and then returns control to the calling point.

For example, area = Area(5); calls the function Area with the value 5 and stores the returned result in area.

Answer the following: (i) What is the purpose of a function argument? (ii) How many (maximum) values can a function return using return statement? (iii) When is a function executed, and where should a function prototype and function definition appear in a source program? (iv) Write three advantages of functions.

The purpose of a function argument is to pass data from the calling function to the called function.

A function can return at most one value through the return statement.

A function is executed when it is called; its prototype appears before main(), and its definition appears after main() or before it.

Three advantages of functions are: programs become easy to understand, code is reused instead of repeated, and errors are easy to find and remove.

Write a program that call two functions Draw_Horizontal and Draw_Vertical to construct a rectangle. Also write functions Draw_Horizontal to draw two parallel

horizontal lines, and the function Draw_Vertical to draw two parallel vertical lines.

```
#include <stdio.h>
void Draw_Horizontal(void)
{
printf("-------------------\n");
printf("-------------------\n");
}
void Draw_Vertical(void)
{
printf("| |\n");
printf("| |\n");
}
int main(void)
{
Draw_Horizontal();
Draw_Vertical();
Draw_Horizontal();
return 0;
}
```

The two horizontal lines form the top and bottom of the rectangle, and the vertical lines form its sides.

Write a program that prompts the user for the Cartesian coordinates of two points (x1, y1) and (x2, y2) and displays the distance between them. To compute the distance, write a function named Distance() with four input parameters. The function Distance() uses the following distance formula to compute the distance and return the result to the calling function: distance = square root of ((x2 - x1)^2 + (y2 - y1)^2).

```
#include <stdio.h>
#include <math.h>
double Distance(double x1, double y1, double x2, double y2)
{
return sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));
}
int main(void)
{
double x1, y1, x2, y2;
printf("Enter x1 y1 x2 y2: ");
scanf("%lf %lf %lf %lf", &x1, &y1, &x2, &y2);
printf("Distance = %f\n", Distance(x1, y1, x2, y2));
return 0;
}
```

For the points (0, 0) and (3, 4), the distance is 5, verified with gcc.

Write a program that prompts the user to enter a number and then reverse it. Write a function Reverse to reverse the number. For example, if the user enters 2765, the function should reverse it so that it becomes 5672. The function should accept the number as an input parameter and return the reversed number.

```
#include <stdio.h>
int Reverse(int num)
{
int rev = 0;
while (num > 0)
{
rev = rev * 10 + num % 10;
num /= 10;
```

```
}
return rev;
}
int main(void)
{
int n;
printf("Enter a number: ");
scanf("%d", &n);
printf("Reversed = %d\n", Reverse(n));
return 0;
}
```

For the input 2765, the function returns 5672, verified with gcc.

Write a function named Draw_Asterisks that will print asterisks (*) according to the pattern shown in the following and make a function call from the function main to print the asterisks pattern. Pattern: 10 asterisks, 8 asterisks, 6 asterisks, 4 asterisks, 2 asterisks.

```
#include <stdio.h>
void Draw_Asterisks(void)
{
int i, j;
for (i = 10; i >= 2; i -= 2)
{
for (j = 1; j <= i; j++)
printf("*");
printf("\n");
}
}
```

```
int main(void)
{
Draw_Asterisks();
return 0;
}
```

The pattern decreases by two asterisks in each row: 10, 8, 6, 4, 2.

(Note: the last two rows read as 3 and 1 asterisks in the faint scan; the decreasing-by-two pattern of the first three rows gives 4 and 2, which is used here.)

Write a function Is_Prime that has an input parameter i.e num, and returns a value of 1 if num is prime, otherwise returns a value of 0.

```
#include <stdio.h>
int Is_Prime(int num)
{
int i;
if (num < 2) return 0;
for (i = 2; i * i <= num; i++)
{
if (num % i == 0) return 0;
}
return 1;
}
```

Write a complete C program that inputs two integers and then prompts the user to enter his/her choice. If the user enters 1 the numbers are added, for the choice of 2 the numbers are divided, for the choice of 3 the numbers are multiplied, for the choice of 4 the numbers are divide (divide the larger number by the smaller number, if the denominator is zero display an error message), and for the choice of 5 the program should exit. Write four functions Add(), Subtract(), Multiply() and

Divide() to complete the task.

```
#include <stdio.h>
int Add(int a, int b) { return a + b; }
int Subtract(int a, int b) { return a - b; }
int Multiply(int a, int b) { return a * b; }
float Divide(int a, int b)
{
if (b == 0) { printf("Error: denominator is zero\n"); return 0; }
if (a > b) return (float)a / b;
return (float)b / a;
}
int main(void)
{
int a, b, choice;
printf("Enter two integers: ");
scanf("%d %d", &a, &b);
printf("Enter choice (1-5): ");
scanf("%d", &choice);
switch (choice)
{
case 1: printf("%d\n", Add(a, b)); break;
case 2: printf("%d\n", Subtract(a, b)); break;
case 3: printf("%d\n", Multiply(a, b)); break;
case 4: printf("%f\n", Divide(a, b)); break;
case 5: break;
}
return 0;
```

```
}
```

Note: the question text says choice 2 is 'divided' and choice 4 is also division; the four functions are used as Add for 1, Subtract for 2, Multiply for 3 and Divide for 4, with 5 to exit.

Write a program that prompts the user to enter a number and calls a function Factorial() to compute its factorial. Write the function Factorial() that has one input parameter and returns the factorial of the number passed to it.

```
#include <stdio.h>
long Factorial(int n)
{
long f = 1;
int i;
for (i = 1; i <= n; i++) f *= i;
return f;
}
int main(void)
{
int n;
printf("Enter a number: ");
scanf("%d", &n);
printf("Factorial = %ld\n", Factorial(n));
return 0;
}
```

Write a function GCD that has two input parameters and returns the greatest common divisor of the two numbers passed to it. Write a complete C program that inputs two numbers and call the function GCD to compute the greatest common divisor of the numbers entered.

```
#include <stdio.h>
int GCD(int a, int b)
{
int t;
while (b != 0) { t = a % b; a = b; b = t; }
return a;
}
int main(void)
{
int a, b;
printf("Enter two numbers: ");
scanf("%d %d", &a, &b);
printf("GCD = %d\n", GCD(a, b));
return 0;
}
```

Chapter 14: File Handling in C

Exercise 14c — Question 1 (Fill in the blanks — book key, p. 242)

A text file can store text only.

EOF stands for End of File .

The fopen function is used to open a file.

An opened file must be closed before terminating the program.

A file opened in a+ mode can be read and appended.

A file pointer is a variable of type FILE* .

A pointer is a memory location whose contents points to another memory location.

In C, every valid string ends with a null character (\0) .

A string is an array of characters.

The fopen() returns a NULL , if it fails to open a file for some reason.

Exercise 14c — Question 2 (Multiple choice — book key, p. 242)

A file is stored in:

A. RAM

✓ B. hard disk

C. ROM

D. cache

Which of the following mode open only an existing file for both reading and writing:

A. "w"

✓ B. "w+"

C. "r+"

D. "a+"

Which of the following functions is used to write a string to a file?

A. puts()

B. putc()

✓ C. fputs()

D. fgets()

On successfully closing a file, the fclose() returns:

A. NULL

✓ B. 0 (zero)

C. 1 (one)

D. FILE pointer

An array subscript should be:

✓ A. int

B. float

C. double

D. an array

Exercise 14c — Question 3 (True / False — book key, p. 242)

A picture can not be stored in a text file

True

EOF marks the end of a string

False

A null character marks the end of a text file

False

Text files are stored in a FILE* (file pointer)

False

The name of the array points to its first element

True

Array subscript is used to access array elements

True

An array of characters can store data of any data type

False

A binary file is a group of contiguous memory locations

False

C can handle text files only

False

When an existing file is opened in 'w' mode, its contents are over-written

True

Exercise 14c — Questions 4-10 (Descriptive and programs)

Can a file be used for both input and output by the same program?

Yes, a file can be used for both input and output by the same program.

A file opened in 'r+' mode can be read and written, and a file opened in 'a+' mode can be read and appended.

In this way one program can read data from a file and also write new data into the same file.

What is a stream? Illustrate the difference between text and binary streams.

A stream is a flow of data between a program and a file or device.

A text stream stores data as characters, so a text file can store text only and can be read by a person.

A binary stream stores data in the same form as it is in memory, so it can store pictures and other non-text data as well.

How many modes are there for opening a file in C? Discuss characteristics of different file opening modes.

The main file opening modes in C are r, w, a, r+, w+ and a+.

The 'r' mode opens an existing file for reading, 'w' creates or over-writes a file for writing, and 'a' appends data to a file.

The 'r+' mode opens an existing file for reading and writing, 'w+' creates a file for reading and writing, and 'a+' opens a file for reading and appending.

What is a file pointer? Briefly explain the concept.

A file pointer is a variable of type FILE*, which points to the information about an opened file.

When a file is opened with fopen(), the function returns a FILE pointer, which is stored in a pointer variable.

All later operations on the file, such as reading, writing and closing, are done through this file pointer.

Write a program to merge the contents of two text files.

```
#include <stdio.h>
int main(void)
{
FILE *f1, *f2, *fout;
int ch;
f1 = fopen("file1.txt", "r");
f2 = fopen("file2.txt", "r");
fout = fopen("merged.txt", "w");
if (f1 == NULL || f2 == NULL || fout == NULL) { printf("Error opening files\n");
return 1; }
while ((ch = fgetc(f1)) != EOF) fputc(ch, fout);
while ((ch = fgetc(f2)) != EOF) fputc(ch, fout);
fclose(f1); fclose(f2); fclose(fout);
return 0;
}
```

The program copies every character of the first file and then of the second file into the merged file.

Write a program that counts the total number of characters in a text file. [Note: consider the blank space a character].

```
#include <stdio.h>
int main(void)
{
FILE *fp;
int ch, count = 0;
fp = fopen("data.txt", "r");
if (fp == NULL) { printf("File not found\n"); return 1; }
while ((ch = fgetc(fp)) != EOF) count++;
```

```
fclose(fp);
printf("Total characters = %d\n", count);
return 0;
}
```

Every character read, including blank spaces, increases the count by one, until EOF marks the end of the file.

Write a program that counts the number of words in a text files and display the count on the screen.

```
#include <stdio.h>
int main(void)
{
FILE *fp;
int ch, words = 0, inWord = 0;
fp = fopen("data.txt", "r");
if (fp == NULL) { printf("File not found\n"); return 1; }
while ((ch = fgetc(fp)) != EOF)
{
if (ch == ' ' || ch == '\n' || ch == '\t') inWord = 0;
else if (inWord == 0) { inWord = 1; words++; }
}
fclose(fp);
printf("Total words = %d\n", words);
return 0;
}
```

A new word is counted whenever a non-space character follows a space, new line or tab.

Appendix: Arrays (preserved from the old notes — flagged: this book has NO Arrays chapter)

Preserved old selection — Arrays (flagged: no Arrays chapter in this book's Contents, pp. 240-242 are Answers/Glossary/Index)

What is an array?

An array is a collection of elements of the same data type, stored in contiguous memory locations under one name.

Each element is reached by its subscript, which is an integer starting from 0.

For example, int marks[5]; stores five integers, reached as marks[0] to marks[4].

What are the advantages of arrays?

Arrays store many values of the same type under a single name, so fewer variables are needed.

Elements are reached quickly by subscript, and loops can process all elements easily.

Arrays also use contiguous memory, which makes processing of lists and tables simple and fast.